

Topic to discuss

- Numerical differentiation using stirling formula
- Numerical problem

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Numerical differentiation using Stirling Central difference formula:

$$y(x) = y_0 + u \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2-1)}{3!} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{u^2(u^2-1)}{4!} \Delta^4 y_{-2} + \dots$$

where, $u = \frac{x - x_0}{h}$, $x = x_0 + uh$

differentiating y w.r.t u ,

$$h y'(x) = \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + u \Delta^2 y_{-1} + \left(\frac{3u^2-1}{6} \right) \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \left(\frac{2u^3-u}{12} \right) \Delta^4 y_{-2} + \dots$$

$$y'(x_0 + uh) = \frac{1}{h} \left[\left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + u \Delta^2 y_{-1} + \frac{3u^2-1}{6} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \left(\frac{2u^3-u}{12} \right) \Delta^4 y_{-2} + \dots \right]$$

$$y'(x_0+uh) = \frac{1}{h} \left[\left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + u \Delta^2 y_{-1} + \frac{3u^2-1}{6} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \left(\frac{2u^3-u}{12} \right) \Delta^4 y_{-2} + \dots \right]$$

Again differentiating w.r.t u ,

$$y''(x) = \frac{1}{h^2} \left[\Delta^2 y_{-1} + u \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \left(\frac{6u^2-1}{12} \right) \Delta^4 y_{-2} + \dots \right]$$

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Numerical problem:

Use Stirling formula and find y' and y'' at $x=0$ from the table.

x	-2	-1	0	1	2
y	4	1	0	1	4

Solution:

$$u = \frac{x - x_0}{h}$$
$$= \frac{0 - 0}{1} = 0$$

$$\therefore u = 0$$

stirling central difference table,

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
$x_{-2} = -2$	$y_{-2} = 4$	$y_{-2} = -3$	$y_{-2} = 2$	$y_{-2} = 0$	$y_{-2} = 0$
$x_{-1} = -1$	$y_{-1} = 1$	$y_{-1} = -1$	$y_{-1} = 2$	$y_{-1} = 0$	
$x_0 = 0$	$y_0 = 0$	$y_0 = 1$	$y_0 = 2$		
$x_1 = 1$	$y_1 = 1$	$y_1 = 3$			
$x_2 = 2$	$y_2 = 4$				

Using stirling y' formula,

$$y'(x_0 + uh) = \frac{1}{h} \left[\left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + u \Delta^2 y_{-1} + \frac{3u^2 - 1}{6} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{u(2u^2 - 1)}{12} \Delta^4 y_{-2} + \dots \right]$$

$$y'(x) = \frac{1}{h} \left[\left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) - \frac{1}{6} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) \right]$$

$$= \frac{1}{1} \left[\frac{1 + (-1)}{2} - \frac{1}{6} \times 0 \right]$$

$$y' = 0$$

$\therefore y'$ at $x = 0$ is 0 Ans.

Now, using double derivative,

$$y'' = \frac{1}{h^2} \left[\Delta^2 y_{-1} + u \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{6u^2 - 1}{12} \Delta^4 y_{-2} + \dots \right]$$

$$= \frac{1}{h^2} \left[\Delta^2 y_{-1} - \frac{1}{12} \Delta^4 y_{-2} \right]$$

$$= \frac{1}{1} \left[2 - \frac{1}{12} \times 0 \right]$$

$$= 2$$

So, y'' at $x=0$ is 2 Ans.

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